

03. CHRONOLOGY OF THE DIFFERENT SYSTEMS

DURATION : 07:33

STUDY SHEET

COURSE SUMMARY

This video explores the timeline of communication systems within cells, highlighting the different stages from autocrine communication to the creation of complex systems. It reveals how the cell, through autocrine communication, becomes aware of its state, and then interacts with its neighbours via paracrine communication. As the systems evolve, structures such as the primitive digestive tube, the connective, circulatory, and endocrine systems emerge, leading to the development of the first organs, notably the primitive heart, and a complexification of the nervous systems. This improved understanding helps osteopathic practitioners grasp the interconnectedness between the different layers of the organism.

THE EVOLUTION OF CELLULAR COMMUNICATION: AN EMBRYOLOGICAL PERSPECTIVE

In **biodynamic osteopathy**, understanding the **chronology of appearance** of different communication systems is fundamental. This approach illuminates how organisms develop and adapt, from the **primitive cell** to the **complex human being**.

AUTOCRINE COMMUNICATION: KNOWING ONESELF

The very first form of communication to appear is **autocrine communication**.

Imagine a small cell. Around it, there is an environment, a membrane, and another environment. What does this cell do? It ejects a fluid outside itself.

Rapidly, this cell develops a small receptor on its membrane, called a **glycocalyx**, which is actually sugar. This receptor allows it to retrieve the fluid it has ejected and reincorporate it.

This means the cell is informed about itself. It's almost philosophical: before you can interact with the outside world, you need to know what you are.

It's a bit like the exercise we did this morning: taking three minutes to settle, connect with your body, your breath, your mind, and observe your state. Recognising this state is a form of autocrine communication, but at a cellular level. This is the first step.

PARACRINE COMMUNICATION: EXCHANGE WITH THE NEIGHBOUR

Next, the system evolves to develop **paracrine communication**.

It is based on the same principle as autocrine communication. I have two cells: one cell emits a small fluid, and the neighbouring cell, right next to it, receives this fluid.

This is paracrine communication, "next to". It is the fastest communication that exists in the body and is continuously used in your **homeostasis** system.

THE EMERGENCE OF SYSTEMS: FROM NOURISHMENT TO COMPLEXITY

The system continues to evolve. The fundamental idea is the **constant search for nourishment**.

I have one cell, then a group of cells. They function well thanks to autocrine and paracrine communications. In a group, we are stronger, more efficient.

THE PRIMITIVE DIGESTIVE TUBE

Primarily, to be efficient in the search and assimilation of nourishment, a **tube** develops. This is an adaptation pattern. I have created a system to distribute information when it arrives.

This is called the **primitive digestive tube**.

THE CONNECTIVE SYSTEM

It is only after the appearance of a primitive digestive system that a network, the **connective system**, develops between my cells.

THE CIRCULATORY SYSTEM

And it is only when I have a connective network that I will develop a **circulatory system**, in order to further improve exchanges.

THE ENDOCRINE SYSTEM

It is only when I have a circulatory system that I will be able to develop an **endocrine system**. Why? Because the cell can release a substance that will travel into the connective system, then into the circulatory system, and be distributed at a distance.

THE FIRST ORGANS AND THE CARDIAC SYSTEM

After functioning well in my network, I start looking for more specific things. I create **first organs**.

One of the very first organs to develop is the **primitive heart**. I develop within my network a system to distribute better, more quickly, a contractile system.

THE PRIMITIVE NEURO-ENTERIC SYSTEM

Then, a system called the **primitive neuro-enteric system** appears. This means I will develop a system where information enters, I need to close, then afterwards I will need to open: this is the **parasympathetic** system.

Then I will start to become more complex, by developing an **orthosympathetic** system.

MOVEMENT AND SENSES

Next, I will need to improve my **movement** in space. In this movement, I will need to develop a **sensory system**.

Then only develop a much more centred, **skeletal** system, and finally my limbs.

CHRONOLOGY AND CLINICAL IMPLICATIONS

This chronology of appearance is almost superimposed between **phylogenesis** (the evolution of species) and **ontogenesis** (the development of the individual).

For example, a child presenting with a motor development problem: it is essential to examine what happens before. Many practitioners, faced with this, work on the cranial. However, by observing the chronology, it is often necessary to go back to the digestive, connective, or circulatory system.

This implies that your **first brain** is enteric, it is not at all here (in the head). The cephalic brain developed logically, initially for itself. We have developed it so much that we have lost sight of this connection, we are cut off.

This chronology is important and must be learned in this sense. We will see that in ontogenetic development, this will follow, even if we do not always see it outwardly.

- le système métabolique autocrine.
- Le système fluide (multicellulaire) paracrine.
- Le système digestif.
- Le système conjonctif.
- Le système circulatoire.
- Le système endocrinien.
- Le système organique.
- Apparition du système neuro-crane.
 - Le système neuronal entérique
 - Le parasympathique
 - L'orthosympathique
- Le système neural moteur.
- Le système sensitif.
- Le système squelettique central.
- les membres.

FIG. — Biological and anatomical systems